Chapter Two

Proportion, Profit and Loss

In our daily life we face various types of problems and we can easily solve all these problems using the concept and explanation of ratio and proportion. So, students should have the concept of ratio and proportion and they must acquire the skill of their application. Similarly, we come across at large scale with transactions and its related loss and profit in our daily life. In this context students should have clear knowledge about profit and loss. In this chapter the topics related to ratio, proportion and profit-loss are discussed in detail.

At end of this chapter, the students will be able to –

- Explain the ratio of multiple expressions and the successive ratios.
- Explain the concept of proportion.
- Solve problems related to proportions.
- Explain what profit and loss is.
- Solve problems related to profit and loss.
- Solve problems of daily life related to tax, VAT and money exchange.
- Solve problems related to time and work, tube and tank, time and distance and boat and tide using unitary method and ratio in real life.

2·1 Ratio of multiple expressions and successive ratios

Ratio of multiple expressions: Let us suppose that the length, breadth and height of a box are 8 cm, 5 cm. and 6 cm. respectively.

The ratio of length, breadth and height = 8 : 5 : 6
In brief, length : breadth : height = 8 : 5 : 6

Here, the ratio of three quantities is represented. The ratio of three or more quantities of this type is called the ratio of multiple expressions.

Successive ratio : Suppose that the ratio of ages of son and father = 15 : 41 (antecedent: subsequent) and the ratio of the ages of father and grandfather = 41 : 65. When two ratios are put together, we get son's age : father's age : grandfather's age = 15 : 41 : 65.

This type of ratio is called successive ratio. It is to be noted that the subsequent of first ratio which is the age of the father in this case is equal to antecedent of second ratio. If the subsequent of first ratio is not equal to the antecedent of second ratio, then they are made equal to find the successive ratio.

To convert two ratios into successive ratio, both the antecedent and the subsequent of second ratio are to be multiplied by the subsequent of first ratio and both antecedent and subsequent of first ratio are to be multiplied by antecedent of second ratio.

Example 1. 7 : 5 and 8 : 9 are two ratios. Express them as successive ratio.

Solution : First ratio = 7 : 5

$$= \frac{7}{5}$$

$$= \frac{7 \times ⑧}{5 \times ⑧} = \frac{56}{40}$$

$$= 56 : 40$$

Second ratio $= 8 : 9$

$$= \frac{8}{9}$$

$$= \frac{8 \times ⑤}{9 \times ⑤} = \frac{40}{45}$$

$$= 40 : 45$$

Alternative solution :

first ratio $= 7 : 5 = 7 \times ⑧ : 5 \times ⑧$

$= 56 : 40$

second ratio $= 8 : 9 = 8 \times ⑤ : 9 \times ⑤$

$= 40 : 45$

∴ Successive ratio of two ratios is 56 : 40 : 45

Activity : Express the following ratios as successive ratio :

1. 12 : 17 and 5 : 12
2. 23 : 11 and 7 : 13
3. 19 : 25 and 9 : 17
4. 5 : 8 and 12 : 17

2·2 Proportion

Suppose that Shohag bought a packet of chips with Tk. 10 and 1 kg. salt with Tk. 25 from a shop. Here the ratio of the prices of salt and chips = 25 : 10 or 5:2.

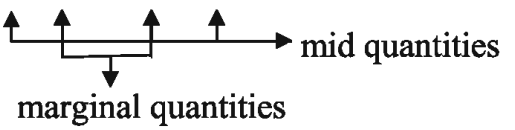
Again, there are 70 students in the class of Shohag. Among them, there are 50 boys and 20 girls. Here the ratio of the number of boys and girls = 50 : 20 or 5 : 2. In both the cases, the ratios are equal.

Therefore, we can say, 25 : 10 = 50 : 20. There are 4 quantities in this ratio. These 4 quantities have made one proportion. Among them if we consider first quantity 25, second quantity 10, third quantity 50 and fourth quantity 20, then we can write,

first quantity: second quantity = third quantity : fourth quantity.

> Of the four quantities, if the ratio of first and second quantity and the ratio of third and fourth quantity are equal to each other, then the four quantities form a proportion. Each quantity of proportion is called proportional.

The first and second quantities of a proportion will be of same kind and the third and fourth quantities will be of same kind. Hence it is not necessary that the four quantities should be of same kind. If two quantities of each ratio are of same kind, then a proportion will be formed. the first and fourth quantities of proportion are called marginal quantities and second and third quantities are called mid-quantities. In proportion the symbol ': :' is used instead of the symbol '='. So, we can write 25 : 10 : : 50 : 20.



Again, first quantity : second quantity = third quantity : fourth quantity

$$\text{or} \quad \frac{\text{first quantity}}{\text{second quantity}} = \frac{\text{third quantity}}{\text{fourth quantity}}$$

or first quantity × fourth quantity = second quantity × third quantity

The rule of three

We know, first quantity × fourth quantity = second quantity × third quantity
Suppose that, first, second and third quantities are 9, 18 and 20 respectively.

Then, $9 \times \text{fourth quantity} = 18 \times 20$

$$\therefore \quad \text{fourth quantity} = \frac{\cancel{18}^{2} \times 20}{\cancel{9}_{1}} = 40$$

$\therefore$ The fourth quantity is 40

In this way, if the three quantities are known, then fourth quantity can be determined. The method of finding the fourth quantity is called **rule of three.**

Observe,

- first and fourth quantities of proportion are called marginal quantities.
- second and third quantities of proportion are called mid-quantities.

Example 2. Determine the fourth proportional of 3, 6, 7.

Solution : Here, the first quantity is 3, the second quantity is 6 and the third quantity is 7

We know that,

first quantity × fourth quantity = second quantity × third quantity

$\therefore$ $3 \times$ fourth quantity $= 6 \times 7$

or fourth quantity $= \frac{\overset{2}{\cancel{6}} \times 7}{\underset{1}{\cancel{3}}}$ or, 14

The required fourth proportional is 14

Example 3. Determine the third quantity of 8, 7 and 14.

Solution : Here, the first quantity is 8, the second quantity is 7 and the fourth quantity is 14.

We know that, first quantity × fourth quantity = second quantity × third quantity

or, $8 \times 14 = 7 \times$ third quantity

$\therefore$ third quantity $= \frac{8 \times \overset{2}{\cancel{14}}}{\underset{1}{\cancel{7}}} = 16$

The required third quantity is 16

Activity : Fill in the gaps of the following :

(a) ☐ : 9 :: 16 : 8

(b) 9 : 18 :: 25 : ☐

Continued Proportion

Let us suppose that, we can form two ratios 5 : 10 and 10 : 20 with three quantities Tk. 5, Tk. 10 and Tk. 20. Here, we can write 5 : 10 : : 10 : 20
This type of proportion is called **continued proportion.** Tk. 5, Tk. 10 and Tk. 20 are called **continued proportional**.

Out of three quantities if the ratio of first and second quantities and the ratio of second and third quantities are mutually equal, then the proportion is called continued proportion. The three quantities are called continued proportional.

If three quantities *a, b, c* are proportionals of the proportion $a : b : : b : c$, then $\frac{a}{b} = \frac{b}{c}$ or, $a \times c = b^2$. That is, multiplication of first and third quantities is equal to the square of second quantity.

Observe : • Second quantity is called mid-proportional or mid-quantity of first and third quantities.

• The three quantities of continued proportion are of same kind.

Example 4. If first and third quantities of a continued proportion are 4 and 16, determine mid-proportional and continued proportion.

Solution : We know that, first quantity × third quantity = (second quantity)^2
Here, first quantity = 4 and third quantity = 16.

$\therefore$ $4 \times 16 = (\text{mid-quantity})^2$

or, $(\text{mid-quantity})^2 = 64$

$\therefore$ $\text{mid-quantity} = \sqrt{64} = 8$

The required continued proportion is 4 : 8 :: 8 : 16 and mid-proportional is 8.

Example 5. If the price of 5 notebooks is Tk. 200. What is the price of 7 notebooks?

Solution : Here, if the number of notebooks is increased, the price of notebooks will be increased.
That is, the ratio of the number of notebooks = the ratio of the prices of notebooks

$\therefore$ 5 : 7 = Tk. 200 : the price of 7 notebooks

or, $\frac{5}{7} = \frac{\text{Tk. } 200}{\text{the price of 7 notebooks}}$

$$\text{or, the price of 7 notebooks} = \frac{7 \times \text{Tk. } 200}{5} = \text{Tk. } 280.$$

Example 6. 12 persons can do a work in 9 days. At the same rate of working, how many days would 18 persons take to complete the work?

Solution : It is to be noted that, if the number of persons be increased, then the period of time will be reduced. Again, if the period of time is to be reduced, the number of persons will have to be increased.

Here, the simple ratio of the number of persons is equal to the inverse ratio of time

$$12 : 18 = \text{The required time} : 9 \text{ days}$$

$$\text{or, } \frac{12}{18} = \frac{\text{Required time}}{9 \text{ days}}$$

$$\text{or, The required time} = \frac{2 \times 9}{3} \text{ days} = 6 \text{ days}$$

Proportional division

Suppose that Tk. 500 is to be distributed in the ratio of 3 : 2.

Here, the sum of antecedent and subsequent of the ratio $3 : 2 = 3 + 2 = 5$

$$\therefore \text{ The first portion} = \frac{3}{5} \text{ parts of Tk. } 500 = \text{Tk. } 300$$

$$\text{and the second portion} = \frac{2}{5} \text{ parts of Tk. } 500 = \text{Tk. } 200$$

So, quantity of one part = given quantity $\times \frac{\text{propotional number of that part}}{\text{sum of the antecedent and subsequent quantities}}$

In the this way a quantity may be divided into various parts.

To divide a given quantity into various parts of fixed ratio is called proportional division.

Example 7. 20 m. cloth is to be divided in the ratio 5 : 3 : 2 among three siblings Amit, Sumit and Chaitee. What is the quantity of cloth for each ?

Solution : Quantity of cloth = 20 m.

given ratio = 5 : 3 : 2

Sum of the numbers of ratio = 5 + 3 + 2 = 10

$\therefore$ The part of Amit $= \frac{5}{10}$ parts of 20 m. = 10 m.

The part of Sumit $= \frac{3}{10}$ parts of 20 m. = 6 m.

and the part of Chaitee $= \frac{2}{10}$ parts of 20 m. = 4 m.

The part of Amit, Sumit and Chaitee are 10m, 6m and 4m respectively.

Activity :

1. If a : b = 4 : 5, b : c = 7 : 9, determine, a : b : c.
2. If Tk. 4800 is divided in the ratio 4 : 3 : 1 among Aisha, Feroja and Khadija, then how much will each of them get?
3. Tk. 570 are to be divided among three students in the ratio of their ages. If their ages are 10, 13 and 15 years respectively, then how much will each of them get?

Example 8. The ratio of income of Ponir and Tapon is 4 : 3, and that of Tapan and Robin is 5 : 4. If the income of Ponir is Tk. 120, what is the income of Robin ?

Solution: The ratio of income of Ponir and Tapan is $4 : 3 = \frac{4}{3} = \frac{4\times5}{3\times5} = \frac{20}{15} = 20:15$

The ratio of income of Tapon and Robin is $5 : 4 = \frac{5}{4} = \frac{5\times3}{4\times3} = \frac{15}{12} = 15 : 12$

Ponir's income : Tapon's income : Rabin's income = 20 : 15 : 12

$\therefore$ Ponir's income : Robin's income = 20 : 12

or, $\frac{\text{Ponir's income}}{\text{Robin's income}} = \frac{20}{12}$

or, Robin's income $= \frac{\text{Ponir's income}\times12}{20}$

$= \text{Tk. } \frac{\cancel{120}^{6}\times12}{\cancel{20}_{1}} = \text{Tk. } 72$

$\therefore$ Robin's income is Tk. 72

Exercise 2·1

1. Write down proportion using the following quantities :

   (a) 3 kg, Tk. 5, 6 kg. and Tk. 10

   (b) 9 years, 10 days, 18 years and 20 days

   (c) 7 cm. 15 seconds, 28 cm. and 1 minute

   (d) 12 notebooks, 15 pencils, Tk. 20 and Tk. 25

   (e) 125 boys and 25 teachers, Tk. 2500 and Tk. 500

2. Two marginal quantities of the continued proportion are given below. Form the proportion:

   (a) 6, 24 (b) 25, 81 (c) 16, 49 (d) $\frac{5}{7}, 1\frac{2}{5}$ (e) 1·5, 13·5.

3. Fill in the gaps :

   (a) 11 : 25 :: ☐ : 50 (b) 7 : ☐ :: 8 : 64 (c) 2·5 : 5·0 :: 7 : ☐

   (d) $\frac{1}{3} : \frac{1}{5}$ :: ☐ : $\frac{7}{10}$ (e) ☐ : 12·5 :: 5 : 25

4. Determine the fourth proportional of the following quantities:

   (a) 5, 7, 10 (b) 15, 25, 33 (c) 16, 24, 32

   (d) $8, 8\frac{1}{2}, 4$ (e) 5, 4·5, 7

5. If the price of 15 kg rice is Tk. 600, what is the price of 25 kg rice?

6. 550 shirts are made daily in a garments factory. How many shirts are made at the same rate in a week?

7. Ages of three sons of Mr. Kabir are 5 years, 7 years and 9 years respectively. He gave Tk. 4200 in the ratio of their ages. How much will each of them get ?

8. If Tk. 2160 is divided among Rumi, Jesmin and Kakali in the ratio of 1 : 2 : 3, how much will each of them get ?